IHARQ Cumulative Evidence Map Through P02

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> **Primary deliverable.** This Evidence Map is the current navigational/provenance layer between governed scientific interpretation and downstream Layer 10 rendering. It organizes evidence; it does not approve claims. Claim authorization is inherited from Layer 0.

1. Provenance model

```
Claim -> Finding -> Interpretation/Analysis -> Metric/Statistic
-> Protocol/Comparison -> Run -> Execution Artifact -> Source/External
Artifact -> Manifest/Hash/Revision
```

Limitations, negative evidence, historical supersessions, literature roles, figures and tables are linked into the same chain. External literature is never treated as project empirical evidence.

2. Coverage summary

- Reviewed claim rows: **37** (19 preserved P00/P01 + 10 P02 + 8 cumulative).
- Finding rows: **49**.
- P02 canonical analysis-input evidence nodes: **38**.
- Total evidence nodes: **69**.
- Artifact registry rows: **47**.
- Relationship rows: **146**.
- Current Layer 10 figures: **14**; tables: **13**.

3. Historical preservation

The P00/P01 Evidence Map is not rebuilt from memory. Its 19 claim rows are preserved from `IHARQ-CUMULATIVE-EVIDENCE-MAP-THROUGH-P01-R1`, and the predecessor package is embedded byte-for-byte under `preserved_prior/`. Additive P02/cumulative relationships do not change historical Layer 0 wording or disposition.

4. P02 evidence saturation

The finalized P02 Phase Analysis had **104/104 expected-evidence requirements dispositioned** (72 fully analyzed, 32 not applicable), **46 actual evidence families dispositioned**, and **38 canonical analysis inputs**. This downstream map preserves those evidence families through claim, finding, limitation, negative-evidence or non-claim supporting roles. No material P02 evidence family is permitted to disappear merely because it is not selected for a Layer 10 figure.

P02 claim-to-evidence mappings

P02-CLM-001/v1 Layer 0 disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed claim: Under the frozen P01 subject-grouped binary motor-imagery protocol, P02 established a complete multi-family raw accept-all decoder/baseline measurement spine across the three activated public EEG datasets; this is Layer-2 measurement evidence and does not establish calibration, abstention, clinical effectiveness, safety, or deployment readiness.

Findings: `P02-FIND-001`

Run: `P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d`

Analysis sources: `analysis_inputs/a0_completion.json`, `analysis_inputs/a0_participant_metrics.c`

Exact numeric support: `{"A0_terminal_cells": 678, "SUCCESS": 657, "FAILED": 3, "CONDITIONAL_SKIP": 15, "DEPENDENCY_BLOCKED": 3, "datasets": 3}`

Limitations: `PUBLIC_EEG_ONLY`, `NO_CALIBRATION_POLICY_CLAIM`

Layer 10: figures none; tables `L10-P02-TBL-001`

P02-CLM-002/v1 Layer 0 disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed claim: Within full-training A0, descriptive decoder ordering was dataset-dependent: CSP-LDA led the single-participant BNCI2014_001 test split, while DBConformer had the highest participant-first mean BACC on Lee2019_MI and PhysioNetMI. These rankings are descriptive and do not establish a universal winning decoder.

Findings: `P02-FIND-002`, `P02-FIND-003`

Run: `P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d`

Analysis sources: `table_sources/P02_Table_A0_fulltrain_branch_summary.csv`

Exact numeric support: `{"full_training_descriptive_leaders": {"BNCI2014_001": {"branch_id": "CLS-CSP-LDA", "participant_first_mean_BACC": 0.6388888888888888, "participants": 1}, "Lee2019_MI": {"branch_id": "DNN-SEQ", "participant_first_mean_BACC": 0.795, "participants":`

6}, "PhysioNetMI": {"branch_id": "DNN-SEQ", "participant_first_mean_BACC": 0.649724517906336, "participants": 11}}}

Limitations: DATASET_HETEROGENEITY, BNCI_TEST_PARTICIPANTS_1

Layer 10: figures L10-P02-FIG-001, L10-P02-FIG-009, L10-P02-FIG-010; tables L10-P02-TBL-001

P02-CLM-003/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: On PhysioNetMI full-training A0, the governed Friedman comparison detected overall decoder heterogeneity (statistic 20.544, p=0.00451, Kendall W=0.267), while no individual alternative-versus-CSP-LDA contrast survived Holm correction; the omnibus result therefore does not identify a statistically supported pairwise winner.

Findings: P02-FIND-003

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/a0_statistics.json

Exact numeric support: {"PhysioNetMI_Friedman_statistic": 20.544, "p_value": 0.00451, "Kendall_W": 0.267, "Lee2019_MI_Friedman_p": 0.148, "Holm_significant_alt_vs_CSP_LDA": 0}

Limitations: MULTIPLICITY, PARTICIPANT_HETEROGENEITY

Layer 10: figures L10-P02-FIG-001; tables L10-P02-TBL-001

P02-CLM-004/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: Across the governed 1, 2, 4, 8, 16 and 32 labels/class P02 low-label surface, performance was difficult, non-monotonic and rank-changing under one frozen subset per budget; the evidence does not support a stable or universal sample-efficiency ordering among the evaluated branches.

Findings: P02-FIND-005

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/low_label_curve_summary.json, analysis_inputs/low_label_metric_source.csv

Exact numeric support: {"governed_low_label_budgets_per_class": [1, 2, 4, 8, 16, 32], "leader_changes": {"BNCI2014_001": 4, "Lee2019_MI": 4, "PhysioNetMI": 6}, "design": "single frozen subset per budget"}

Limitations: SINGLE_FROZEN_SUBSET, NO_LOW_LABEL_REPEAT_STABILITY

Layer 10: figures L10-P02-FIG-002, L10-P02-FIG-003, L10-P02-FIG-004; tables L10-P02-TBL-002, L10-P02-TBL-003

P02-CLM-005/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: The preserved P02 training-correction history shows that model viability and optimization behavior under the implemented configurations depended materially on training-policy choice. This supports treating inadequate optimization as distinct from demonstrated architectural incapacity, but does not establish universal test-set gains, architecture superiority, or theoretical invalidity of the earlier generic policy.

Findings: P02-FIND-006, P02-FIND-008

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: runtime/diagnostics/runtime_successor/stage11_maximum_analysis/stage11_sour
machine_readable P02 Protocol run matrix

Exact numeric support: {"Stage12_validation_only_examples": [{"dataset_id": "BNCI2014_001", "branch_id": "DNN-FBCNET", "A_BACC": 0.5, "B_BACC": 0.7222222222222222, "absolute_delta_BACC": 0.2222222222222222, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "DNN-FBCNET", "A_BACC": 0.6395092565383486, "B_BACC": 0.6601038299539622, "absolute_delta_BACC": 0.0205945734156136, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "DNN-SEQ", "A_BACC": 0.6985502987559996, "B_BACC": 0.7167940052894505, "absolute_delta_BACC": 0.0182437065334508, "claim_bearing": false}, {"dataset_id": "PhysioNetMI", "branch_id": "SSL-CBRAMOD", "A_BACC": 0.5, "B_BACC": 0.6384072876873348, "absolute_delta_BACC": 0.1384072876873348, "claim_bearing": false}], "test_role_loaded": false}

Limitations: POST_OBSERVATION_AMENDMENT, DIAGNOSTIC_VALIDATION_EVIDENCE

Layer 10: figures L10-P02-FIG-005; tables L10-P02-TBL-004, L10-P02-TBL-005

P02-CLM-006/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: The Stage-11 S&R diagnostic challenger did not yield a consistent participant-level BACC improvement over the canonical primary EEGNet across BNCI2014_001, Lee2019_MI and PhysioNetMI; it remains diagnostic-only and does not replace primary A0 evidence.

Findings: P02-FIND-007

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/training_policy_challenger_statistics.json

Exact numeric support: {"diagnostic_challenger": [{"dataset_id": "BNCI2014_001", "n": 5, "median_delta_BACC": -0.0347222222222222, "ci_low": -0.0763888888888889, "ci_high": -0.0347222222222221, "p_value": 0.125, "rank_biserial": -0.8666666666666667}, {"dataset_id":

"Lee2019_MI", "n": 30, "median_delta_BACC": 0.01, "ci_low": -0.005, "ci_high": 0.025, "p_value": 0.2309220325143355, "rank_biserial": 0.2586206896551724}, {"dataset_id": "PhysioNetMI", "n": 55, "median_delta_BACC": 0.0, "ci_low": -0.0343477914915631, "ci_high": 0.0029761904761905, "p_value": 0.455185419014922, "rank_biserial": -0.1190130624092888}]}

Limitations: DIAGNOSTIC_ONLY

Layer 10: figures L10-P02-FIG-005; tables L10-P02-TBL-005

P02-CLM-007/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: Within the canonical resource-constrained A4 scope, none of the 756 governed C1-C3 role-control statistical rows reached raw p<=0.05. This is an absence of statistically detected improvement under the executed scope, not proof that the true effect is zero and not evidence of five-repeat or dense-budget stability.

Findings: P02-FIND-009, P02-FIND-010

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/a4_completion.json, analysis_inputs/stage18_resource_constra

Exact numeric support: {"governed_C1_C3_statistical_rows": 756, "raw_p_le_0_05": 0, "scope": "single canonical refit repeat / governed deep anchor budgets"}

Limitations: SINGLE_REPEAT, DEEP_ANCHOR_BUDGETS

Layer 10: figures L10-P02-FIG-006; tables L10-P02-TBL-006

P02-CLM-008/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: A hard-vote ensemble was not uniformly beneficial: one governed PhysioNetMI full-training C4 comparison at repeat 4 showed a Holm-significant disadvantage relative to the validation-selected EEGNet constituent (median delta BACC -0.0435; Holm p=0.0371). This cell-specific negative result does not establish that ensembles are generally inferior.

Findings: P02-FIND-011

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/a4_c4_c5_statistics.json

Exact numeric support: {"Holm_significant_rows": 1, "significant_rows": [{"comparison_id": "P02-A4-PhysioNetMI-ENSEMBLE-A4-C4-MODEL-HARD-VOTE-FULL_TRAIN-ER04:VS_STF "dataset_id": "PhysioNetMI", "condition_id": "A4-C4-MODEL-HARD-VOTE", "budget_id": "FULL_TRAIN", "repeat": 4, "strongest_constituent_branch": "DNN-EEGNET", "closure_status": "COMPLETE", "n": 11.0, "median_delta_BACC":

-0.0434782608695651, "p_value": 0.0185546875, "holm_adjusted_p": 0.037109375, "rank_biserial": -0.7878787878787878, "ci_low": -0.1304347826086956, "ci_high": -0.0119047619047618}]}

Limitations: CELL_SPECIFIC_EFFECT

Layer 10: figures none; tables L10-P02-TBL-007

P02-CLM-009/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: Post-hoc Stage18S sensitivity evidence shows that A4 effect direction can change across descriptive repeats and budgets (22/45 anchor trajectories mixed-sign; all 15 six-budget MR00 trajectories changed sign at least once). This supports the existing claim ceiling but remains POST_HOC/DESCRIPTIVE/SENSITIVITY evidence and does not modify canonical Stage18/G18.

Findings: P02-FIND-012

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/stage18S_R1_decision_support_summary.json, analysis_inputs/stage18S_R1_three_repeat_anchor_stability.csv, analysis_inputs/stage18S_R1_six_budget_mr00_sensitivity.csv

Exact numeric support: {"three_repeat_anchor_trajectories": 45, "all_negative": 7, "mixed": 22, "all_positive": 16, "six_budget_trajectories": 15, "six_budget_with_sign_flip": 15}

Limitations: POST_HOC_ONLY, THREE_REPEAT_NOT_FIVE_REPEAT

Layer 10: figures L10-P02-FIG-007; tables L10-P02-TBL-008

P02-CLM-010/v1 Layer 0 disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed claim: P02 provides a fail-closed, provenance-rich Layer-2 evidence substrate for downstream use while preserving successful and non-success outcomes, including 648 FailureCaseIndex records, 340 NegativeResultNotes and 309 DiagnosticOnlyFlags. This is technical evidence readiness, not deployment readiness or future-phase scientific success.

Findings: P02-FIND-013, P02-FIND-014

Run: P02-L2-OFFICIAL-RUN-R4DYN-9e181d2e935d

Analysis sources: analysis_inputs/failure_negative_summary.json, negative_and_failed_results/

Exact numeric support: {"FailureCaseIndex": 648, "NegativeResultNote": 340, "DiagnosticOnlyFlag": 309, "failure_terminal_breakdown": {"CONDITIONAL_SKIP": 591, "NOT_APPLICABLE_REPEAT_SLOT": 42, "INPUT_INCOMPATIBLE": 9, "FAILED": 3, "DEPENDENCY_BLOCKED": 3}}

Limitations: DOWNSTREAM_SCIENCE_NOT_EXECUTED

Layer 10: figures L10-P02-FIG-008; tables L10-P02-TBL-009

5. Cumulative P00-P02 claim provenance

CUM-CLM-001/v1

Disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed text: Through P02, IHARQ has a governed evidence chain linking validated foundation contracts, the frozen P01 provenance-traceable EEG substrate, and P02 Layer-2 decoder evidence through explicit identities, manifests and handoffs; this traceability does not by itself establish clinical or deployment effectiveness.

Cumulative findings: CF-01, CF-02, CF-09

Contributing phase evidence: resolved through cumulative_findings.yaml, results_to_claims.yaml, and the phase-local finding/evidence nodes.

Limitations: Technical traceability is not scientific/clinical effectiveness by itself.

CUM-CLM-002/v1

Disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed text: The P02 decoder analyses consume the frozen P01 subject-grouped, denominator-conserving binary motor-imagery data contract rather than a P02-local redefinition of labels, split or core windows, subject to the scope of the implemented registered integrity and leakage checks.

Cumulative findings: CF-02

Contributing phase evidence: resolved through cumulative_findings.yaml, results_to_claims.yaml, and the phase-local finding/evidence nodes.

Limitations: Bounded to the implemented registered leakage/disjointness checks.

CUM-CLM-003/v1

Disposition: APPROVED_WITH_QUALIFICATIONS

Reviewed text: Across the three governed P02 datasets, decoder-family ordering is context-dependent rather than universally ordered, and participant heterogeneity on the multi-participant datasets limits any global winner claim; BNCI2014_001 is descriptive at the participant-inference level because its held-out P02 test set contains one participant.

Cumulative findings: CF-03, CF-11

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: BNCI held-out test participant n=1; no universal rank claim.

CUM-CLM-004/v1

Disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed text: The deterministic low-label populations prepared in P01 expose difficult, non-monotonic and rank-changing P02 behavior across the governed 1-32 labels/class surface; one frozen subset per budget prevents a stable repeated-subset sample-efficiency claim.

Cumulative findings: `CF-04`

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: Single frozen subset per budget; no repeated-subset stability.

CUM-CLM-005/v1

Disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed text: P01 resolved A4 data-feasibility by freezing a fully matched R2 substrate. Under P02's accepted constrained A4 execution, no generic C1-C3 statistically detected improvement was observed, and post-hoc Stage18S further shows repeat/budget sign sensitivity. These results do not establish zero effect, five-repeat stability, dense-budget behavior, or full original-grid equivalence.

Cumulative findings: `CF-05`, `CF-06`

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: Single canonical refit repeat, deep anchor budgets, Stage18S post-hoc.

CUM-CLM-006/v1

Disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed text: The P02 correction history shows that apparent model viability under the implemented configurations depended on training-policy choice while the inherited P00/P01 data and identity foundations remained unchanged; this supports a reproduction/optimization-sensitivity interpretation, not universal architecture superiority or guaranteed gains from source-aligned settings.

Cumulative findings: `CF-07`

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: Diagnostic/validation evidence does not establish universal test-set gains or architecture superiority.

CUM-CLM-007/v1

Disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed text: Across P00-P02, fail-closed validation and explicit retention of failures, supersessions, null/negative results and diagnostic states provide an auditable project history in which non-success evidence remains available for interpretation; this process does not guarantee scientific correctness or deployment safety.

Cumulative findings: `CF-08`, `CF-09`

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: Different negative-evidence classes are not quantitatively interchangeable.

CUM-CLM-008/v1

Disposition: `APPROVED_WITH_QUALIFICATIONS`

Reviewed text: At the end of P02, canonical Layer-2 evidence is technically ready for governed downstream consumption, including P03, while explicit scope, repeat/stability, access and non-clinical limitations remain mandatory handoff constraints; this does not imply future-phase scientific success.

Cumulative findings: `CF-12`

Contributing phase evidence: resolved through `cumulative_findings.yaml`, `results_to_claims.yaml`, and the phase-local finding/evidence nodes.

Limitations: Downstream science is not yet executed.

6. Negative and superseded evidence

- P01 A4 effectiveness claim `P01-CLM-011/v1` remains historically DEFERRED and is linked to later P02 A4 evidence rather than rewritten.
- P02 retains 648 FailureCaseIndex records, 340 NegativeResultNotes and 309 DiagnosticOnlyFlags as distinct evidence classes.
- Stage11 diagnostic challenger remains diagnostic-only.
- Stage18S remains post-hoc descriptive sensitivity evidence.
- Failed, dependency-blocked, incompatible, skipped and not-applicable cells remain visible and mapped.

7. External artifact provenance

P01 numerical arrays remain governed by their Kaggle pointer identities. P02 uses a separate Hugging Face credential class. The preferred P02 continuation source is the immutable full-workspace snapshot; the chunked release is archival/fallback. Mutable `latest` retrieval is prohibited.

Credential classes: `IHARQ_HF_TOKEN_PRE_P02` and `IHARQ_HF_TOKEN_P02`. Literal values are never stored.

Preferred P02 continuation source

- repository: `Csthv/p02-phase02-final-whole-working-20260814t052120z-b890ff92`
- immutable revision: `bc14961e14f2e48690e55df3577014275f9cbf30`
- manifest: `__IHARQ_SNAPSHOT__/WORKING_TREE_MANIFEST.jsonl`
- manifest SHA-256: `564bbb44e67d14da8457ad83d33651ded93dfc6b89ab90f10f78a36a20b8e8f5`
- per-object integrity index SHA-256: `540a0c3d4cbc686645ab1fccf088205c02c32a5c34e4b238c91f9c4ec`

Archival P02 source

- repository: `Csthv/iharq-p02-phase02-r2-P02-L2-OFFICIAL-RUN-R4DY-20260814t051352z`
- immutable final release revision: `257a407b5c7ea37b6c620863ee261c010c8f197c`
- content revision: `bc88dccd6518aa7b172697be1ef98344997f27ff`
- manifest revision: `c60c0505faa8076090439277ed1d8ef8c86734b7`

8. Literature evidence boundary

Literature is represented with explicit roles such as `EXTERNAL_METHOD_REFERENCE` and `EXTERNAL_CONTEXTUAL_COMPARISON`. It may contextualize EEGNet, FBCNet, DBConformer, CBraMod, the public datasets and BCI reproducibility, but it cannot prove an IHARQ measured result. Bibliographic metadata was freshly verified on 2026-08-14.

9. Figure/table two-way provenance

Every current Layer 10 figure/table is registered in `figure_table_provenance.yaml` with claim IDs, finding IDs, source data, output path, checksum and limitations. No Layer 10 artifact is allowed to create a new scientific population, metric, denominator or claim.

10. Orphan tests

- Reviewed claims without findings/evidence: **0 material orphans**.
- P02 material evidence without finding/limitation/negative/support role: **0 material orphans**.
- Broken P02 external revision/hash references: **0 identified**.
- Claim-bearing Layer 10 artifacts without Evidence Map support: **0**.

EVIDENCE_MAP_THROUGH_P02_STATUS = FINALIZED